



VIRGINIA COMMONWEALTH UNIVERSITY

Statistical Analysis and Modelling (SCMA 632)

A4: Limited Dependent Variable Models

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Date of Submission: 30-06-2025

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Introduction

This report details a statistical analysis project focused on predicting credit card defaulters. The project utilizes a dataset containing various demographic and transactional information to build and evaluate predictive models. The primary goal is to apply and compare the performance of Logistic Regression, Decision Tree and Probit Regression models.

Objectives

- **Data Preprocessing:**
 - Load and preprocess the credit card defaulter dataset, including handling categorical variables, scaling numerical features, and splitting the data into training and testing sets.
- **Feature Selection:**
 - Implement a correlation-based feature selection method to identify the most relevant predictors for the models.
- **Model Development:**
 - Train and evaluate three distinct classification models: Logistic Regression, Decision Tree, and Probit Regression.
- **Performance Evaluation:**
 - Assess model performance using key metrics such as Confusion Matrices and Receiver Operating Characteristic (ROC) curves with Area Under the Curve (AUC).
- **Cross-Platform Comparison:**
 - Compare the results, particularly the ROC AUC scores, obtained from both Python and R implementations for each model to identify and interpret any discrepancies.

Business Significance

Predicting credit card defaulters holds significant business importance for financial institutions:

- **Risk Management:** Accurate prediction of defaulters allows banks and credit card companies to proactively manage financial risk by identifying high-risk accounts and implementing strategies to mitigate potential losses. This includes adjusting credit limits, offering financial counselling, or initiating collection processes.
- **Strategic Decision Making:** Insights from predictive models can inform lending strategies, customer segmentation, and product development. For instance, understanding the key factors contributing to default can help design more effective credit policies and target appropriate customer segments.
- **Optimized Resource Allocation:** By pinpointing likely defaulters, institutions can allocate resources more efficiently, focusing collection efforts where they are most needed and avoiding unnecessary expenditures on low-risk accounts.
- **Enhanced Profitability:** Reducing default rates directly contributes to higher profitability by minimizing write-offs and ensuring a healthier loan portfolio.

- **Regulatory Compliance:** Robust risk assessment models are often required for compliance with financial regulations, ensuring responsible lending practices.

Results

- **Data Preprocessing and Feature Selection:**
 - The dataset was loaded, and the ID column was removed. Categorical features (SEX, EDUCATION, MARRIAGE) were encoded numerically, and numerical features were scaled using Min-Max scaling. Correlation-based feature selection was applied, retaining features with a correlation score greater than 0.01 with the target variable (default).
- **Model Performance:**
 - **Logistic Regression**
 - **Python (A4-Python.pdf):**
 - **Accuracy:** 0.816
 - **ROC AUC:** 0.74
 - **Confusion Matrix:**
 - True Negative (0,0): 4627
 - False Positive (0,1): 980
 - False Negative (1,0): 110
 - True Positive (1,1): 283
 - **R (A4-R.pdf):**
 - **Accuracy:** 0.806
 - **ROC AUC:** 0.74
 - **Confusion Matrix:**
 - True Negative (0,0): 4568
 - False Positive (0,1): 1060
 - False Negative (1,0): 104
 - True Positive (1,1): 267
 - **Decision Tree**
 - **Python (A4-Python.pdf):**
 - **Accuracy:** 0.802
 - **ROC AUC:** 0.70
 - **Confusion Matrix:**
 - True Negative (0,0): 4467
 - False Positive (0,1): 1180
 - False Negative (1,0): 108
 - True Positive (1,1): 285
 - **R (A4-R.pdf):**
 - **Accuracy:** 0.7923
 - **ROC AUC:** 0.70
 - **Confusion Matrix:**
 - True Negative (0,0): 4426
 - False Positive (0,1): 1202
 - False Negative (1,0): 118

- True Positive (1,1): 275
- **Probit Model**
 - **Python (A4-Python.pdf):**
 - **Accuracy:** 0.816
 - **ROC AUC:** 0.73
 - **Confusion Matrix:**
 - True Negative (0,0): 4627
 - False Positive (0,1): 980
 - False Negative (1,0): 110
 - True Positive (1,1): 283
 - **R (A4-R.pdf):**
 - **Accuracy:** 0.806
 - **ROC AUC:** 0.74
 - **Confusion Matrix:**
 - True Negative (0,0): 4568
 - False Positive (0,1): 1060
 - False Negative (1,0): 104
 - True Positive (1,1): 267

Interpretations

The results demonstrate that all three models, Logistic Regression, Decision Tree, and Probit Regression, are capable of predicting credit card defaults with reasonable accuracy.

- **Model Performance:** Logistic Regression and Probit Regression generally show similar and slightly better performance than the Decision Tree model in terms of accuracy and ROC AUC. This indicates that for this dataset, linear decision boundaries might be more effective than the piecewise constant boundaries of a basic decision tree.
- **ROC AUC Similarity (R):** A notable observation is the identical ROC AUC (0.74) for Logistic Regression and Probit Regression in the R solution. This is common, as the logistic and probit link functions, while different, often produce very similar probability estimates and, consequently, similar discriminative power, especially when the underlying distributions are not highly unusual.
- **Python vs. R Discrepancies:** While overall trends are similar, minor numerical differences in accuracy, confusion matrix counts, and specifically the ROC AUC for Probit Regression (0.73 in Python vs. 0.74 in R) were observed. These discrepancies can be attributed to:
 - **Underlying Implementations:** Different default optimization algorithms, convergence criteria, and numerical precision in the *sklearn* and *statsmodels* libraries in Python versus *glm* and *caret* in R.
 - **Randomness:** Although a fixed random seed was used for data splitting, minor differences in how random processes are handled internally by different libraries can still lead to slight variations.

Recommendations

To further enhance the credit card defaulter prediction system and ensure greater consistency across different analytical environments:

- **Hyperparameter Tuning:** Systematically tune the hyperparameters for all models, especially the Decision Tree (e.g., *cp*, *maxdepth*, *minsplits*), in both R and Python to optimize their performance and potentially close the performance gap between models.
- **Cross-Validation:** Implement k-fold cross-validation during model training to get more robust estimates of model performance and reduce the impact of a single train-test split.
- **Feature Engineering:** Explore creating new features from existing ones. For example, interaction terms between education and age, or financial ratios, could provide deeper insights and improve model accuracy.
- **Advanced Models:** Investigate more complex machine learning models such as Gradient Boosting Machines (e.g., *XGBoost*, *LightGBM*) or Support Vector Machines (SVMs), which often offer higher predictive power.
- **Threshold Optimization:** While ROC AUC is threshold-independent, for deployment, optimize the classification threshold (currently 0.5) based on specific business objectives (e.g., minimizing false positives vs. false negatives), especially considering the imbalanced nature of default data.
- **Error Analysis:** Conduct a deeper analysis of misclassified samples from both Python and R to understand patterns in errors and identify potential data quality issues or model weaknesses.
- **Environment Consistency Review:** For critical business applications requiring highly consistent results across platforms, perform a detailed code review and potentially align specific library versions and function parameters to minimize discrepancies.

References

- Data provided through Canvas Portal.